

Functions of Several Variables & Partial Derivatives



Evaluating functions of two variables

- 1 For $f(x, y) = x^2y - 3xy + y^2$, find $f(2, -1)$.
 - 2 For $f(x, y) = \sqrt{x^2 + y^2}$, find $f(3, 4)$ and describe the domain of f .
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Computing partial derivatives

- 3 For $f(x, y) = x^4y^3 + 2xy - y^5$, find f_x and f_y .
 - 4 For $f(x, y) = e^{x^2y}$, find f_x and f_y .
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Second-order and mixed partial derivatives

- 5 For $f(x, y) = x^3y^2$, find f_{xx} , f_{yy} , and f_{xy} .
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Clairaut's Theorem: equality of mixed partials

- 6 For $f(x, y) = \sin(xy) + x^2y$, verify that $f_{xy} = f_{yx}$.
 - 7 State Clairaut's Theorem and the condition it requires.
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Domains and level curves

- 8 Find the domain of $f(x, y) = \ln(x - y)$.
 - 9 Describe the level curve $f(x, y) = 4$ for $f(x, y) = x^2 + y^2$.
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More partial derivative practice

- 10 For $f(x, y) = x^3 - 4xy^2 + y^3$, find f_x and f_y .
 - 11 For $f(x, y) = \ln(x^2 + y^2)$, find f_x and f_y .
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Functions of three variables

- 12 For $f(x, y, z) = x^2yz^3$, find f_x , f_y , and f_z .
- 13 For $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$, find $f(1, 2, 2)$.

More second-order partials practice

14 For $f(x, y) = e^{xy}$, find f_{xx} , f_{yy} , and f_{xy} .

Level curves as a family of curves

15 Describe the family of level curves $f(x, y) = c$ for $f(x, y) = y - x^2$, as c varies.

Domains involving multiple restrictions

16 Find the domain of $f(x, y) = \frac{\sqrt{9 - x^2 - y^2}}{y}$.

One more round of partial derivative practice

17 For $f(x, y) = \frac{x}{y}$, find f_x and f_y .

18 For $f(x, y) = \cos(x + 2y)$, find f_x and f_y .

A closing pair: evaluation and partials together

19 For $f(x, y) = x^2 e^y$, find $f(2, 0)$, $f_x(2, 0)$, and $f_y(2, 0)$.

20 Describe the level surface $f(x, y, z) = 1$ for $f(x, y, z) = x^2 + y^2 + z^2$.

A final pair on mixed partials

21 For $f(x, y) = x^2 \sin y$, find f_{xy} and f_{yx} , and confirm they match.

22 For $f(x, y) = \frac{y}{x}$ ($x \neq 0$), find f_{xx} .

Sketching a family of level curves

23 Sketch (describe) the level curves $f(x, y) = c$ for $f(x, y) = y - x^3$, for $c = -1, 0, 1$.

24 For the same $f(x, y) = y - x^3$, find f_x and f_y , and verify ∇f is never 0.

25 For $f(x, y, z) = xyz + x^2$, find f_x , f_y , f_z , and the second-order partial f_{xz} .

Piecing together a full evaluation problem

26 For $f(x, y) = x^2 + 2xy - y^2 + 3x - 4$, find $f(1, -2)$, $f_x(1, -2)$, and $f_y(1, -2)$.

27 For the same $f(x, y) = x^2 + 2xy - y^2 + 3x - 4$, find f_{xx} , f_{yy} , and f_{xy} .

A boundary-behavior question

28 Find the domain of $f(x, y) = \sqrt{x} + \sqrt{y}$, and describe what happens to f as (x, y) approaches a boundary point like $(0, 4)$.